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PROTECTIVE ELEMENT, SET OF COMPONENTS AND METHOD FOR PRODUCING THE PROTECTIVE ELEMENT

Abstract

A protective element for arrangement on a device which emits electromagnetic radiation, wherein the device can be, for example, an at least partially electrically powered motor vehicle, the protective element comprising the following: a shielding layer for shielding against electromagnetic radiation, a plastic layer, and a contacting zone via an electrical contact can be established between the shielding layer and the device.

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Background/Summary

RELATED APPLICATION [0001] This application is a continuation of international application No. PCT/EP2023/082416 filed on Nov. 20, 2023, and claims the benefit of German application No. 10 2022 130 663.1 filed on Nov. 21, 2022, which are incorporated herein by reference in their entirety and for all purposes.

FIELD OF DISCLOSURE

[0002] The present invention pertains to the field of protective elements, more particularly of EMC protective elements.

[0003] Electromagnetic compatibility (EMC) denotes the absence of interference to electrical or electronic apparatuses caused by its environment.

BACKGROUND

[0004] Modern motor vehicles are at least partially electrically powered. Furthermore, increasing numbers of devices that communicate by way of electromagnetic waves are being integrated into motor vehicles or utilized in motor vehicles. Securing the EMC is consequently becoming a growing challenge.

[0005] In at least partially electrically powered motor vehicles, particularly in the region around the battery, there is additionally a requirement to ensure a protective function against external effects such as stone impact, and to ensure integrity in the event of thermal runaway in battery cells.

[0006] This is to be made possible with minimal component weights, so that the weight of the motor vehicle is increased only to a minor extent or not at all.

SUMMARY OF THE INVENTION

[0007] It is an object of the present invention to provide an effective protective element at low cost and complexity. The protective element preferably may be (as far as possible) universally employable for the securement of the EMC and/or may offer protection against stone impact and/or protection against thermal runaway in a battery cell. It may be particularly advantageous if the protective element is advantageous overall, taking account of production cost and complexity, securement of the EMC, protection against stone impact and protection against thermal runaway in battery cells.

[0008] The object is achieved in accordance with the invention by the features of claim 1.

[0009] The device which emits electromagnetic radiation may be, for example, an at least partially electrically powered motor vehicle.

[0010] The device, for example the at least partially electrically powered motor vehicle, preferably comprises an electrochemical unit for providing electrical energy serving to propel the motor vehicle. The electrochemical unit may preferably be a battery unit or a fuel cell unit.

[0011] The device, for example the at least partially electrically powered motor vehicle, preferably comprises one or more electrical machines which serve to propel the motor vehicle.

[0012] Within the device, such as within the at least partially electrically powered motor vehicle, the electromagnetic radiation may be emitted by the electrochemical unit and/or by the one or more

electrical machines or else by other components of the device.

[0013] The protective element comprises a shielding layer for shielding from electromagnetic radiation. Shielding layers suitable for shielding from electromagnetic radiation are familiar to those skilled in the art from the technical field of electromagnetic compatibility.

[0014] The thickness of the shielding layer may be, for example, 0.01% to 50% of the wall thickness of the protective element. In the case in particular of protective elements of low surface area, which are subject to little mechanical stress, even a very thin shielding layer may account for a high proportion of 50%, for example, of the wall thickness of the protective element. Conversely, in the case of protective elements of very high surface area, which may be subject to severe mechanical stress, as in the case of underbody protective elements, for example, a substantial wall thickness may be rational. In that case it may oftentimes be completely sufficient for the shielding layer to account only for a very low proportion of 0.01%, for example, of the wall thickness of the protective element. It may be preferable if the thickness of the shielding layer is 0.5% to 20% of the wall thickness of the protective element. It may be particularly preferable if the thickness of the shielding layer is 0.8% to 15% of the wall thickness of the protective element.

[0015] The thickness of the shielding layer may advantageously be 0.01 to 3 mm. It may be preferable if the thickness of the shielding layer is 0.02 to 1 mm. It may be particularly preferable if the thickness of the shielding layer is 0.03 to 0.3 mm.

[0016] The wall thickness of the protective element may advantageously be 1 mm to 5 cm. It may be preferable if the wall thickness of the protective element is 1.5 mm to 2.5 cm. It may be particularly advantageous if the wall thickness of the protective element is 2 to 8 mm. In the case in particular of protective elements of low surface area, which are exposed only to low mechanical stresses, very low wall thicknesses may be sufficient. Conversely, high wall thicknesses may be particularly advantageous in the case of protective elements of high surface area, which are exposed to relatively high mechanical stresses.

[0017] Materials suitable for shielding layers are familiar to those skilled in the art. The shielding layer may advantageously comprise: a metal; and/or an electrically conductive carbon; and/or a carrier layer comprising an electrically conductive layer.

[0018] The electrically conductive carbon may be, for example, a graphite or a carbon fiber. The carrier layer may be, for example, a plastic film or a plastic nonwoven, e.g. a PET nonwoven. Polyethylene terephthalate is referred to in abbreviated form as PET. The electrically conductive layer may be an aluminum coating, copper coating, NiCo coating or silver coating applied to the carrier layer.

[0019] It is particularly advantageous if the shielding layer comprises a metal. The metal may comprise, for example, aluminum, copper, iron and/or silver.

[0020] The metal may be in any suitable form in the shielding layer.

[0021] The metal in the shielding layer may preferably be present as a metal foil, metal mesh and/or as a metal nonwoven.

[0022] The shielding layer may for example be selected from: [0023] aluminum foils, [0024] aluminum meshes, [0025] stainless steel meshes, [0026] bronze meshes, [0027] copper meshes, [0028] plastic nonwovens (e.g. PET nonwovens), with applied aluminum film, [0029] plastic nonwovens (e.g. PET nonwovens), coated with copper on one side, [0030] unidirectional tapes containing carbon fiber, [0031] nonwovens made from recycled carbon fibers, [0032] graphite foils, [0033] polymers (e.g. polypropylenes) combined with an aluminum mesh.

[0034] It may be particularly preferable if the shielding layer comprises a metal foil.

[0035] It may be especially preferable if the shielding layer comprises aluminum.

[0036] Hence the shielding layer may be, for example, an aluminum foil or an aluminum sheet, with the thickness of the shielding layer being preferably 0.02 to 1 mm, e.g. 0.03 to 0.3 mm.

[0037] In trials it was found surprisingly that aluminum foil enabled much better shielding effectiveness in the frequency range from 0.01 to 1000 MHz than aluminum meshes of comparable

thickness and than copper, bronze or stainless steel.

[0038] It may be preferable if the shielding layer comprises regularly spaced openings.

[0039] The shielding layer may have multiple plies. A first ply, preferably first metal ply, e.g. first aluminum ply, of the shielding layer may preferably comprise regularly spaced openings, and a second ply, preferably second metal ply, e.g. second aluminum ply, of the shielding layer may preferably have regularly spaced openings, in which case a spacing of the openings in the second ply is different than a spacing of the openings in the first ply. This may be achieved, for example, by meshes or grids having different mesh sizes, or by corrugated foils.

[0040] The protective element comprises a plastic layer. The plastic contained in the plastic layer is preferably a thermoplastic.

[0041] The plastic layer and the shielding layer may be connected to one another indirectly or directly. They are connected to one another indirectly if there is at least one further layer between the plastic layer and the shielding layer.

[0042] The plastic layer and the shielding layer preferably form layers of a layered assembly. The protective element is preferably a layered assembly element.

[0043] It may be advantageous if an adhesion promoter is arranged, in the form of an adhesion promoter coating, for example, between shielding layer and plastic layer, for example on a surface of the shielding layer that faces the plastic layer.

[0044] It may be particularly advantageous if the shielding layer is a metal foil or a metal sheet, more particularly is an aluminum foil or an aluminum sheet, and the adhesion promoter is arranged between the shielding layer and the plastic layer on a surface of the shielding layer that faces the plastic layer.

[0045] The adhesion promoter is preferably selected from adhesion promoters which boost the adhesion on the surface of the shielding layer.

[0046] The adhesion promoter is preferably selected from adhesion promoters which boost the adhesion of plastics on the surface of the shielding layer.

[0047] The adhesion promoter is preferably selected from adhesion promoters which boost the adhesion of the plastic of the plastic layer on the surface of the shielding layer.

[0048] The protective element may preferably comprise an electrically insulating protective layer.

[0049] It may be particularly advantageous if the shielding layer is arranged between the plastic layer and the electrically insulating protective layer.

[0050] The statement that a layer is arranged between other layers allows for further interlayers. For example, there may thus be one or more reinforcing layers between the shielding layer and the plastic layer.

[0051] Reinforcing layers are addressed in detail elsewhere herein.

[0052] An effect of the electrically insulating protective layer may be that in the event of malfunction of a battery unit or a battery cell arranged in the vicinity of the shielding layer, no short-circuiting occurs. Those skilled in the art from the field of battery housings are familiar with suitable electrically insulating materials which can be applied as an electrically insulating protective layer.

[0053] The protective element comprises a contacting zone by means of which electrical contact can be produced between the shielding layer and the device. The contacting zone may preferably be a surface of the shielding layer.

[0054] The electrical contact which can be produced between the contacting zone and the device may be a direct electrical contact or an indirect electrical contact between the surface of the shielding layer and the device. If the contact is an indirect contact, the contact may be an electrical contact bridgeable by a bridging element between the shielding layer and the device.

[0055] It may be particularly advantageous if the protective element comprises a contacting element.

[0056] By means of the contacting element, preferably, electrical contact, more particularly direct

or indirect electrical contact, can be producible or stabilizable between the contacting zone and the device.

[0057] By means of the contacting element, for example, direct electrical contact can be producible or stabilizable between the contacting zone and the device.

[0058] In the region of the contacting zone, the contacting element here may have been pressed from an inner region of the protective element onto the shielding layer. As a result, in the region of the contacting zone, the shielding layer may be supported from an inner region of the protective element such that an outer surface of the shielding layer, which may be the contacting zone, can be pressed firmly onto the device. For example, a protrusion element described herein may have been pressed onto the shielding layer, so that the contacting zone at that location (e.g. locally) curves toward the device, which may make it possible to produce more stable electrical contact to the device. The contacting element may in that case be electrically conductive or electrically insulating. Indeed, it may promote the production or stabilization of direct electrical contact between the shielding layer and the device substantially through a supporting effect. The contacting element in that case need not itself conduct electrical current.

[0059] Electrical contact bridged by the contacting element can preferably be producible between the contacting zone and the device. For example, a surface of the contacting element may be in electrically conductive contact with the contacting zone, which may be a surface of the shielding layer, for example, and a further surface of the contacting element may be able to be brought into electrically conductive contact with the device. In that case, the contacting element is electrically conductive.

[0060] It may be particularly advantageous if the contacting element is deformable and/or elastic.

[0061] It may be particularly advantageous if the contacting element comprises fibers and/or a fibrous material.

[0062] The contacting element and/or fibers of the contacting element and/or the fibrous material of the contacting element may advantageously extend wholly or partially around a compression zone. An example of what may be interpreted as a compression zone is a cavity which may extend, for example, in the contacting element. The compression zone and/or the cavity may preferably be compressed and/or squashed in an assembly direction on assembly of the protective element.

[0063] The contacting element may for example be a hollow contacting element through which the compression zone extends. It may be advantageous if the contacting element is a fibrous material hose, e.g. a woven fabric hose.

[0064] Even if it is deformable and/or elastic and/or comprises the fibers and/or the fibrous material, the contacting element of course preferably comprises a sufficiently electrically conductive material, so that electrical contact between the contacting zone and the device can be producible or stabilizable by means of the contacting element.

[0065] It may be advantageous if the protective element comprises a positioning and/or aligning element, which may for example be or comprise a positioning and/or aligning projection.

[0066] At least one portion of the positioning and/or aligning element advantageously can be arranged on at least one portion of the contacting zone.

[0067] It may be advantageous if at least one portion of the contacting element, preferably of the fibrous material hose, e.g. of the woven fabric hose, is arranged on the positioning and/or aligning element, e.g. on the positioning and/or aligning projection, and/or if at least one portion of the contacting element, preferably of the fibrous material hose, e.g. of the woven fabric hose, is arranged on the contacting zone.

[0068] This may be advantageous as the contacting element can thus be held in the desired position in a particularly simple way.

[0069] A fibrous material hose, e.g. a woven fabric hose, may preferably be fluid-permeable through its walls.

[0070] The contacting element may preferably be annular. The annular contacting element may be

a sleeve. A form of this kind may be advantageous, as a connecting element, e.g. a screw, may be guided through the opening in the annular contacting element or in the sleeve in order to secure the protective element on the device.

[0071] Advantageously, at least part of the contacting element may be materially bonded with the plastic layer and/or at least part of the contacting element may be incorporated in the plastic layer. This may be brought about, for example, in a particularly elegant way by positioning the contacting element at the desired point in the shaping process so that a plastic material, e.g. a plastic melt, can spread around the contacting element and incorporate the contacting element into the resultant plastic layer.

[0072] The shielding layer may preferably comprise a shielding layer overhang.

[0073] It may be advantageous if the shielding layer overhang comprises the contacting zone.

[0074] The shielding layer on the shielding layer overhang is preferably not covered by the plastic layer and/or, if the protective element is embedded between the plastic layer and a further layer, e.g. a further plastic layer, the shielding layer is preferably not covered by at least one of these two layers.

[0075] At least part of the shielding layer overhang may extend in a different direction than a part of the shielding layer that is covered by the plastic layer. For example, at least part of the shielding layer overhang may be in contact with an outer lateral surface of the annular contacting element. At least part of the shielding layer overhang may have been pressed onto the lateral surface, have for example been pressed through a surrounding plastic layer.

[0076] The contacting element may comprise a protrusion element.

[0077] The protrusion element may preferably face the contacting zone comprised by the shielding layer, e.g. the contacting zone comprised by the shielding layer overhang.

[0078] The protrusion element may extend through the shielding layer on the contacting zone. It may, for example, penetrate the shielding layer.

[0079] It may be advantageous if the protective element on both sides of the shielding layer comprises respectively at least one contacting element or one part of a contacting element.

[0080] The contacting zone is arranged preferably in an edge zone of the protective element. The edge zone may, for example, adjoin a wall zone of the protective element.

[0081] In the transition from the wall zone into the edge zone, the protective element may have a kink or a curve. Alternatively or additionally, a layer structure of the protective element may be different in an edge zone than in a zone adjoining the edge zone.

[0082] A plurality of contacting zones spaced apart from one another may be arranged in the edge zone.

[0083] A contacting zone extending along the edge zone may be arranged in the edge zone.

[0084] The shielding layer may be wholly or partially embedded. It may be wholly or partially embedded between the plastic layer and a further layer. The further layer may preferably be a further plastic layer, a reinforcing layer, an electrically insulating protective layer and/or a high-temperature thermal insulation layer. The further layer may extend, for example, up to the edge zone, without concealing the edge zone.

[0085] It may be preferable if the electrical contact can be produced between the shielding layer and a frame or a carrier structure comprised by the device. The frame and/or the carrier structure may be comprised by the mass of the device. In the case of motor vehicles, the mass potential is the carrier structure comprising bodywork and frame. The term “mass” used in this context refers to this mass potential.

[0086] It may be preferable if the contacting zone is configured on a shielding layer end face.

[0087] As the shielding layer is in layer form, the shielding layer comprises not only two main surfaces but also an encircling end face.

[0088] It may be particularly advantageous if the shielding layer comprises a passage opening. The passage opening may preferably align with an opening in the annular contacting element. The

opening in the annular contacting element may be the opening which passes through the annular contacting element.

[0089] The passage opening may comprise a shielding layer end face encircling the passage opening, and the contacting zone may be a contacting zone configured on the encircling shielding layer end face, e.g. an inner-surface contacting zone.

[0090] This may be particularly advantageous as a contacting element may be introduced into the passage opening and electrical contact between the shielding layer and the device can be produced via the contacting element. The contacting element may for example be a connecting element, in particular a screw. The thread of the connecting element may engage, for example, into a counterthread in a frame of the device and may thereby ensure electrical contact with the device as well.

[0091] Preferably, a distance of each contacting zone to the most closely adjacent contacting zone is at most 300 mm, preferably at most 200 mm, more preferably at most 100 mm. It may be particularly advantageous if a distance of each contacting zone to the most closely adjacent contacting zone is for example at most 50 mm, preferably at most 30 mm, more preferably at most 20 mm. This is the case especially for distances between inner-surface contacting zones.

[0092] It may be particularly advantageous if the contacting zone is an outer-surface contacting zone, with the outer-surface contacting zone being configured on an outer surface of the shielding layer.

[0093] An outer-surface contacting zone configured on an outer surface of the shielding layer may be present in particular when the shielding layer in the region of the contacting zone forms a capping layer of the protective element.

[0094] A layer referred to herein as a capping layer is always the layer which forms a concluding layer in the layer structure of the protective element. In that case, one of the main surfaces of the capping layer forms one of the main surfaces of the protective element.

[0095] An outer surface of the protective element is understood herein to be, in particular, an outer surface of the protective element that extends parallel to the layered assembly planes of the protective element.

[0096] An outer surface of the shielding layer is understood herein to be, in particular, an outer surface of the shielding layer that extends parallel to the layered assembly planes of the protective element and is preferably also an outer surface of the protective element.

[0097] The protective element may preferably comprise an offset zone. In the offset zone, the shielding layer may preferably extend in a different direction than in the contacting zone.

[0098] It may be advantageous if the shielding layer in the contacting zone forms a capping layer of the protective element. The contacting zone in that case may in particular be an outer-surface contacting zone.

[0099] In a further zone, separated from the contacting zone by the offset zone, a layer of the protective element that is different from the shielding layer may form a capping layer of the protective element. The further zone which is separated from the contacting zone by the offset zone may be an insulating zone, for example.

[0100] The layer of the protective element that is different from the shielding layer and that forms the capping layer of the protective element in the further zone may be, for example, the electrically insulating protective layer described herein.

[0101] This can be particularly advantageous, as in the insulation zone, in the event of abnormal behavior of a battery, in the case of thermal runaway of a battery cell, for example, the electrically insulating protective layer can prevent short-circuits. At the same time, however, in the region of the contacting zone, the shielding layer may be available for electrical contacting. This makes it possible to ensure improved shielding from electromagnetic rays which may be emitted by the battery in regular operation.

[0102] It may be advantageous if the protective element is fiber-reinforced.

[0103] The protective element may be fiber-reinforced, for example, in that the shielding layer, the plastic layer and/or the electrically insulating protective layer comprise fibers.

[0104] It may be advantageous if the plastic layer comprises fibers.

[0105] The fibers may be dispersed in a plastic material of the plastic layer, in which case the plastic material is preferably a thermoplastic material. A mean length of the fibers may be preferably 1 mm to 40 mm, e.g. 2 mm to 20 mm. The fibers may be ground fibers, preferably recycled fibers, e.g. recycled carbon fibers. The plastic material may form a plastic matrix in which the dispersed fibers are distributed.

[0106] It is particularly preferable if the protective element comprises a reinforcing layer.

[0107] The reinforcing layer may preferably comprise a laid fiber scrim or a woven fiber fabric. The laid fiber scrim may be, for example, a uniaxial or a multiaxial scrim.

[0108] The reinforcing layer may be a metallic reinforcing layer, which for example may comprise a further aluminum sheet or a further aluminum foil. In general this is less preferable.

[0109] The plastic layer may be a reinforcing layer. In general this is less preferable. A preferred protective element comprises the reinforcing layer in addition to the plastic layer.

[0110] The plastic layer is preferably thicker than the reinforcing layer. The thickness of the plastic layer may be at least twice the thickness of the reinforcing layer. If a plurality of reinforcing layers are present, the thickness of the plastic layer may be at least twice the thickness of the thickest reinforcing layer.

[0111] The plastic layer may be foamed. Substances which promote foaming of plastics are familiar to those skilled in the art.

[0112] It may be particularly advantageous if the protective element is a protective sandwich element. The protective sandwich element preferably has a respective reinforcing layer on both surfaces. The two reinforcing layers may independently of one another preferably comprise a laid fiber scrim or a woven fiber fabric. The laid fiber scrim may be, for example, a uniaxial or a multiaxial scrim.

[0113] It may be particularly advantageous if the protective sandwich element has at least one respective reinforcing layer on both surfaces of the plastic layer. The two reinforcing layers may independently of one another preferably comprise a laid fiber scrim or woven fiber fabric. The laid fiber scrim may be, for example, a uniaxial or a multiaxial scrim.

[0114] The shielding layer may advantageously run at least in one portion of the protective sandwich element between the plastic layer and a reinforcing layer or between two reinforcing layers which are arranged on one side of the plastic layer.

[0115] It may be particularly advantageous if the protective element comprises a high-temperature thermal insulation layer.

[0116] The term “high-temperature” refers in this context to the temperatures which may arise in the event of battery cell runaway. They may be in a temperature range from 1000 to 1200° C. In the event of thermal runaway of a battery cell, for example, gases with a temperature of about 1100° C. may emanate from the battery cell. In addition to the thermal loading, an abrasive effect may occur. This effect derives from a particle flow. Indeed, together with hot gases, hot particles as well emanate from the battery cell experiencing runaway.

[0117] The high-temperature thermal insulation layer preferably withstands the abrasive stream of hot gas for as long as possible.

[0118] It may be advantageous if the high-temperature thermal insulation layer comprises a hot gas barrier.

[0119] The hot gas barrier may comprise a mineral barrier material.

[0120] The mineral barrier material may be selected from mica and mineral fibers, preferably from glass fibers and ceramic fibers, e.g. from glass fibers.

[0121] Alternatively or additionally, the hot gas barrier may comprise an organic barrier material.

[0122] The organic barrier material may preferably be a meta-aramid material, e.g. a laid meta-

aramid scrim or a woven meta-aramid fabric. Organic barrier materials of these kinds are marketed by DuPont under the brand name Nomex®.

[0123] The organic barrier material may preferably comprise carbon fibers. The organic barrier material may comprise, for example, a regular or an irregular fabric which comprises carbon fibers or is produced from carbon fibers. The regular fabric may preferably be a woven fabric comprising carbon fibers. The irregular fabric may preferably be a nonwoven comprising carbon fibers or a paper comprising carbon fibers. The regular fabric may be, for example, a woven carbon fiber fabric. The irregular fabric may preferably be a carbon fiber nonwoven or a carbon fiber paper.

[0124] It may be particularly advantageous if the hot gas barrier comprises a woven glass fiber fabric.

[0125] The use of a woven glass fiber fabric may be particularly advantageous as lateral blowing of fibers by the hot gas stream is prevented in such a fabric. Lateral blowing of the fibers is prevented in particular by the fibers running transversely in the woven fabric in each case. This structure may considerably retard the passage of the flame and/or of the hot gases.

[0126] It may be advantageous if the mineral barrier material comprises ceramic fibers. The hot gas barrier may comprise, for example, a nonwoven containing ceramic fibers, e.g. a ceramic nonwoven.

[0127] It may be especially advantageous if the hot gas barrier comprises a plurality of mineral barrier materials or comprises a mineral barrier material and an organic barrier material. Hence the hot gas barrier may comprise, for example, a woven glass fiber fabric with mica arranged thereon or a woven glass fiber fabric with additional meta-aramid fiber reinforcement.

[0128] The protective element may be curved.

[0129] The protective element may have one convex and one concave surface. The electrically insulating protective layer and/or the high-temperature thermal isolation layer may preferably form a capping layer of the protective element in a concave region. The electrically insulating protective layer and/or the high-temperature thermal insulation layer may form an inner lining of the protective element. The protective element may in that case serve in particular as a battery housing part, in which case a part of the battery may extend into the concave region of the protective element.

[0130] It may be advantageous if the protective element comprises a protrusion zone, with the protrusion zone protruding from a surface of the protective element and defining the contacting zone. The shielding layer in the protrusion zone preferably protrudes from the surface of the protective element.

[0131] It may be particularly advantageous if the shielding layer in the protrusion zone comprises a bead and the bead protrudes from the surface of the protective element.

[0132] The protective element may comprise an electrically conductive discharge element integrated into the protective element, with the electrically conductive discharge element being in electrically conductive contact with the contacting zone. The electrically conductive discharge element may be, for example, an electrically conductive sleeve. In that case, the contacting zone may in particular be an inner-surface contacting zone encircling a passage opening. On the contacting zone, e.g. on the inner-surface contacting zone which encircles the passage opening, the shielding layer may be concealed with an electrically insulating capping layer. In that case, the electrically conductive discharge element, e.g. the electrically conductive sleeve, may extend at least partially through the electrically insulating capping layer. An electrically insulating capping layer may be, for example, a reinforcing layer.

[0133] It may be preferable if the shielding layer comprises a shielding deflection zone, wherein two portions of the shielding layer overlap in the shielding deflection zone, and one of the two portions forms the contacting zone. The contacting zone may be, for example, an outer-surface contacting zone.

[0134] With preference, one of the two overlapping portions of the shielding layer may extend

along a concave zone of the protective element, and the other of the two overlapping portions of the shielding layer may extend along a convex counterzone of the protective element, with the convex counterzone being arranged in the concave zone.

[0135] The protective element may be obtained at least partially by compression molding. The protective element may in particular be obtained at least partially by the compression molding method described herein.

[0136] The protective element may be obtained at least partially by injection molding. The protective element may in particular be obtained at least partially by an injection molding method described herein.

[0137] The object is also achieved in accordance with the invention by the component set according to the relevant independent claim.

[0138] The component set comprises a contacting element which is in electrically conductive contact with the contacting zone.

[0139] The contacting element is a loose component which is different from the shielding layer and is detachable from the protective element. It differs from the electrically conductive discharge element described herein and integrated into the protective element. The protective element integrated into the electrically conductive discharge element is deemed to be part of the protective element, as it does not constitute an independent component detachable from the protective element.

[0140] The contacting element may preferably be a connecting element. This may be particularly advantageous, as a contacting element which is also a connecting element not only is able to produce electrically conductive contact with the contacting zone but also additionally is able to produce a connection to the mass comprised by the device. The connecting element may be, for example, a screw, a bolt, a nut, a rivet and/or a snap connection element.

[0141] A particularly preferred connecting element is a screw. The screw may in particular be a flow-hole screw.

[0142] The contacting element may be annular. The annular contacting element may preferably be a sleeve.

[0143] It may be advantageous if the contacting element comprises a protrusion element facing the shielding layer. It may be particularly preferable if the contacting element comprises a plurality of protrusion elements facing the shielding layer. The protrusion elements may be arranged, for example, on the end face of a contacting element, more particularly on the end face of a contacting element which is a sleeve. The protrusion elements may be distributed on the end face around an opening which extends through the sleeve. The number of protrusion elements may for example be at least three. The at least three protrusion elements may be distributed on the end face at an equal distance from each other around the opening which extends through the sleeve.

[0144] It may be particularly advantageous if the protrusion element facing the shielding layer mediates electrically conductive contact from an outer surface of the shielding layer to the contacting element when the outer surface of the shielding layer is pressed onto the contacting element.

[0145] The protrusion element may be advantageous as it is able to produce an electrically conductive contact in particular also via passivated surface regions or adhesion promoter located on the surface. This may be advantageous in particular in the production of the protective element, as when attaching adhesion promoters on the shielding layer, for example on the aluminum foil, for instance, the regions of the shielding layer that are intended to form outer-surface contacting zones do not have to be kept free of adhesion promoter. For that it would be necessary, during the application of adhesion promoter, to employ masks or the like which prevent application of adhesion promoter in the regions of later outer-surface contacting zones. That would entail considerable extra cost and complexity.

[0146] On the outer surface of the shielding layer, advantageously, an adhesion promoter may be

arranged and the protrusion element facing the shielding layer may mediate the electrically conductive contact from the outer surface of the shielding layer to the contacting element by perforating the adhesion promoter.

[0147] The protrusion element perforates the adhesion promoters in particular when the outer surface of the shielding layer is pressed onto the contacting element.

[0148] The protrusion element may in particular be an elevation consisting of an electrically conductive material. The elevation may extend with a taper from one surface of the contacting element. The elevation may in particular extend with a taper from one surface of the contacting element and run to a point.

[0149] The contacting element may comprise a protrusion element facing away from the shielding layer. It may be particularly preferable if the contacting element comprises a protrusion element facing the shielding layer and a protrusion element facing away from the shielding layer.

[0150] The contacting element may comprise mutually opposing contacting element surfaces, with at least one respective protrusion element being arranged on the two surfaces.

[0151] The protrusion element facing away from the shielding layer may mediate electrically conductive contact of the contacting element to a mass, e.g. to the bodywork or to the frame of the motor vehicle, when the contacting element is pressed onto the mass. In particular, the protrusion element facing away from the shielding layer may mediate the electrically conductive contact from the contacting element to the mass when the contacting element is inserted between the mass and an outer surface of the shielding layer. The protrusion element facing away from the shielding layer may promote electrically conductive contact also via a means attached on the surface of the mass, in which case the means may be an anticorrosion means, for example. The protrusion element may perforate the means, e.g. the anticorrosion means, and thereby promote the electrically conductive contact.

[0152] The object is also achieved in accordance with the invention by a method according to the relevant independent claim.

[0153] A ply of a shielding material is provided and a plastic layer is formed in the layered assembly with the shielding material. This may mean that the plastic layer is formed directly on the surface of the shielding material. It may also mean that at least one further connecting layer, e.g. a reinforcing layer, is present or is formed between the surface of the shielding material and the plastic layer.

[0154] The shielding material is at least partially reshaped prior to the formation of the plastic layer; and/or at least partially reshaped during the formation of the plastic layer; and/or at least partially reshaped after the formation of the plastic layer in the layered assembly.

[0155] The plastic layer may be produced for example by compression molding, by injection molding or by sprayed application.

[0156] Optionally a further layer is formed in the layered assembly with the shielding material concurrently with the formation of the plastic layer, temporally overlapping the formation of the plastic layer or temporally offset from the formation of the plastic layer.

[0157] Especially if the method comprises a compression molding method, the optional further layer may be formed, for example, from a reinforcing ply and may be a reinforcing layer. In that case, for example, a plastic serving to form the plastic layer may be arranged between the ply of the shielding material and the reinforcing ply.

[0158] Especially if the method comprises an injection molding method, the optional further layer may be, for example, a further plastic layer. The further plastic layer may preferably be formed temporally overlapping the formation of the plastic layer or temporally offset from the formation of the plastic layer.

[0159] A contacting zone is formed by means of which electrical contact can be produced between the shielding layer formed from the shielding material and the device which emits electromagnetic radiation.

[0160] An outer-surface contacting zone, for example, may be formed, wherein an offset zone is formed by reshaping of the shielding material, wherein the shielding layer in the offset zone extends preferably in a different direction than in the outer-surface contacting zone. An inner-surface contacting zone, for example, may be formed. The inner-surface contacting zone may be formed, for example, by the production of a passage opening. The passage opening extends preferably through the shielding layer.

[0161] During the formation of the plastic layer, it is preferably possible for the contacting element, preferably the annular contacting element, the sleeve for example, to be embedded at least partially into the resultant plastic layer, wherein the contacting element may be arranged preferably on a surface of the shielding material at a point where the contacting zone is prelocated or is formed. It may be particularly preferable for the contacting element to be positioned at a desired point and only then for the front of a spreading plastic melt to be conveyed there, so that the plastic spreads around the contacting element and embeds the contacting element.

[0162] It may be advantageous if the contacting zone is formed or modified with a die element and/or a punch element.

[0163] It may be particularly advantageous if the contacting element is held with the die element on the surface of the shielding material, and/or the contacting element is pressed with the die element against the surface of the shielding material. In that case, with the die element, e.g. at least one protrusion element can be pressed against the surface of the shielding material, can be impressed into the surface of the shielding material or can be pressed through the shielding material.

[0164] It may be particularly advantageous if the contacting element is accommodated wholly or partially, preferably partially, in a recess in a mold half of a shaping tool, to be positioned at the desired point.

[0165] It may be advantageous if the passage opening is formed or modified with the punch element.

[0166] It may be particularly preferable if at least one die element and/or at least one punch element is guided in a shaping tool. The at least one die element and/or the at least one punch element may preferably be guided in a mold half of the shaping tool, e.g. in at least one dedicated guiding zone of the shaping tool.

[0167] An effect of this may be that the entire protective element is generated substantially completely in one operation or even in one shaping tool, including precisely positioned contacting elements and exactly defined contacting zones. As a result, ultimately, a particularly effective protective element can be provided with the minimum of cost and complexity.

[0168] The method may comprise a compression molding method, wherein an arrangement which is provided in an open cavity of a shaping tool, e.g. of a compression mold, in a step a) comprises:

[0169] the ply of the shielding material, [0170] a further ply serving to form the further layer, e.g. a reinforcing ply, and [0171] a plastic arranged between these plies and serving to form the plastic layer, e.g. a plastic which is moldable by compression molding.

[0172] The term “plies” is used in connection with the arrangement, since at the moment when the arrangement is provided, in the open cavity of the shaping tool, the plies comprised of the arrangement need not yet be connected to one another, i.e. a layered assembly need not yet be present.

[0173] The shielding material is of course a precursor of the shielding layer. The shielding layer is therefore formed of the shielding material. The ply of the shielding material may therefore have a composition as described herein for the shielding layer.

[0174] The reinforcing ply is a precursor of the reinforcing layer. The reinforcing ply may in particular have a composition as described herein for the reinforcing layer.

[0175] The plastic forms a precursor of the plastic layer. The plastic may in particular have a composition as described herein for the plastic layer.

[0176] In a step b) downstream of the arrangement in the open cavity of the shaping tool, the shape

of the protective element is at least partially defined by closing the shaping tool, by building up an elevated pressure and/or by supplying heat. Here, the shielding material is at least partially reshaped. For that purpose, the elevated pressure and/or the supply of heat are preferably regulated such that the plastic at least partially melts and a layered assembly is formed which comprises: [0177] a shielding layer, [0178] a further layer, e.g. a reinforcing layer, and [0179] a plastic layer arranged between these layers.

[0180] In a step c) downstream of the at least partial definition of the shape of the protective element, the shaping tool is opened and the protective element or a protective element precursor is removed from the open cavity of the shaping tool.

[0181] The protective element precursor may differ from the protective element in that, for example [0182] it does not comprise a contacting zone; and/or [0183] it comprises fewer layers than the protective element.

[0184] After the removal of the protective element precursor from the open cavity of the shaping tool, the protective element precursor may be converted into the protective element by means of at least one subsequent machining step.

[0185] A subsequent machining step may comprise the formation of the contacting zone.

[0186] The contacting zone may be formed by introduction of a passage opening, with the passage opening extending preferably through the shielding layer.

[0187] A subsequent machining step may comprise the attachment of a further layer on the protective element precursor.

[0188] The contacting zone may be formed in the shaping tool.

[0189] The arrangement may preferably further comprise a capping ply on a side, facing away from the plastic, of the ply of the shielding material, wherein the shielding material in the arrangement, in an overhang portion of the arrangement, projects beyond the edge of the capping ply, wherein the action of the elevated pressure and/or the action of the heat produce a displacement force which displaces the shielding material in the overhang portion, to form an offset zone, wherein after the shaping, the shielding material extends in a different direction in the offset zone than in the overhang portion.

[0190] The capping ply may be, for example, the high-temperature thermal insulation layer or a precursor of the high-temperature thermal insulation layer.

[0191] The capping ply may preferably contain a hot-gas barrier. The hot-gas barrier may preferably contain a mineral barrier material. The features, indicated in this connection and described in more detail above in connection with the high-temperature thermal insulation layer, may also be features of the capping ply described in connection with the shaping method.

[0192] It may be particularly advantageous if the arrangement on both sides of the plastic comprises respectively at least one reinforcing ply. It is possible accordingly, in a particularly simple way, to obtain a protective sandwich element which is highly mechanically robust and has a reinforcing layer on both sides of the plastic layer.

[0193] The method may comprise an injection molding method, wherein the ply of the shielding material is positioned between two mold halves of an injection mold in a step a).

[0194] The ply of the shielding material may be interpreted as a precursor of the shielding layer described herein. The features described in connection with the shielding layer may also be features of the shielding material.

[0195] Particularly in connection with the shaping method, e.g. injection molding method, it may be particularly advantageous if the ply of the shielding material is a closed-surface ply. A closed-surface ply refers to a ply which has no passage openings through which, for instance, a fluid could flow through the shielding material, from one surface of the shielding material to the other surface of the shielding material.

[0196] It may be particularly advantageous if the ply of the shielding material is an aluminum foil or an aluminum sheet, wherein the thickness of the ply is preferably 0.02 to 1 mm, e.g. 0.03 to 0.3

mm.

[0197] In a step b) downstream of the positioning of the ply of the shielding material between the mold halves of the injection mold, a shape of the shielding material, desired for the protective element, is defined in the injection mold by reshaping. The shape of the shielding material may be defined in the injection mold by at least partially closing the injection mold. The injection mold may therefore in particular be an integrated injection mold and shaping tool.

[0198] In a step c) downstream of the defining of the desired shape of the shielding material, a plastic layer, e.g. a layer of melted plastic, is applied on at least one of the two surfaces of the shielding material. Features described herein in connection with the plastic layer may be features of the plastic layer applied in this step.

[0199] It may be preferable if in an optional step d) a plastic layer, e.g. a layer of melted plastic, is also applied on the other of the two surfaces of the shielding material. The features described herein in connection with the plastic layer may be features of the plastic layer applied in this step.

[0200] It may be particularly advantageous if in the plastic, e.g. in the plastic which is applied on one of the two surfaces of the shielding material, or in the plastic which is applied on both surfaces of the shielding material, there are fibers dispersed, wherein a mean length of the fibers may be preferably 1 mm to 40 mm, e.g. 2 mm to 20 mm.

[0201] The fibers may be selected from glass fibers, carbon fibers and polymer fibers.

[0202] The method for producing the protective element may be a combined compression molding and injection molding method.

[0203] The method may be a method for producing a protective element, comprising an electrically conductive discharge element integrated into the protective element, wherein the electrically conductive discharge element is in electrically conductive contact with the contacting zone.

[0204] It may be preferable to produce the electrically conductive contact of the discharge element with the contacting zone on the very ply of the shielding material that is introduced into the method for producing the protective element. The discharge element may be integrated into the protective element, for example, by attaching the discharge element on the shielding material through a layer which is formed in the course of the method, for example through the plastic layer, through the electrically insulating protective layer or through the high-temperature thermal insulation layer. The discharge element may be a sleeve which is at least partially surrounded by the layer formed in the course of the method.

[0205] The protective element is preferably a protective element having a concave inside and a convex outside.

[0206] The protective element may be, for example, an underbody protection element. The underbody protection element may be a battery casing floor element intended for arrangement beneath a battery of a motor vehicle.

[0207] The protective element may be a casing element. The casing element may preferably be a casing cover, a casing wall or a casing tray.

[0208] The protective element may preferably be a battery casing element. The battery casing element may preferably be a battery casing cover, a battery casing wall or a battery casing tray.

[0209] The protective element may be a battery casing element for arrangement between the battery and an electrical unit, e.g. an electrical communication unit, of a motor vehicle.

[0210] Further preferred features and/or advantages of the invention are the subject matter of the description below and of the drawings representing exemplary embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0211] FIG. 1 shows a schematic sectional representation of a detail of a protective element;

[0212] FIG. **2** shows a schematic sectional representation of a detail of a protective element;
[0213] FIG. **3** shows a schematic sectional representation of a detail of a protective element;
[0214] FIG. **4** shows an enlarged detail from FIG. **3**;
[0215] FIG. **5** shows section B-B from FIG. **3**;
[0216] FIG. **6** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0217] FIG. **7** shows section A-A from FIG. **6**;
[0218] FIG. **8** shows section B-B from FIG. **6**;
[0219] FIG. **9** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0220] FIG. **10** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0221] FIG. **11** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0222] FIG. **12** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0223] FIG. **13** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0224] FIG. **14** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0225] FIG. **15** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0226] FIG. **16** shows a discharge element;
[0227] FIG. **17** shows a schematic sectional representation of a detail of a protective element, the element being arranged on a mass of a motor vehicle;
[0228] FIG. **18** shows a schematic representation of a shaping method;
[0229] FIGS. **19-26** show illustrative details from FIG. **18** for different layer structures of the resultant protective element;
[0230] FIG. **27** shows an edge portion of a protective element obtainable by the method according to FIG. **18**;
[0231] FIG. **28** shows section A-A from FIG. **27**;
[0232] FIG. **29** shows section B-B from FIG. **27**;
[0233] FIG. **30** shows an illustrative detail from FIG. **18** with a securement arrangement in an overhang portion;
[0234] FIG. **31** shows a schematic representation of the resultant edge portion of a protective element in the context of an overhang portion according to FIG. **30**;
[0235] FIG. **32** shows an edge portion of a protective element, obtainable according to FIGS. **30** and **31**;
[0236] FIG. **33** shows section A-A through FIG. **32** with an additionally represented mass;
[0237] FIG. **34** shows section C-C through FIG. **32** with an additionally represented mass;
[0238] FIG. **35** shows a schematic sectional representation of a shaping tool;
[0239] FIG. **36** shows a shaping tool according to FIG. **35** with arranged shielding material;
[0240] FIG. **37** shows a shaping tool according to FIG. **35** with positioned shielding material;
[0241] FIG. **38** shows a shaping tool according to FIG. **35** with reshaped shielding material;
[0242] FIG. **39** shows a shaping tool according to FIG. **35** after application of a first plastic layer;
[0243] FIG. **40** shows a shaping tool according to FIG. **35** after application of a second plastic layer;
[0244] FIG. **41** shows a shaping tool according to FIG. **40** in a compressed state;
[0245] FIG. **42** shows a shaping tool according to FIG. **41** after a punching step;
[0246] FIG. **43** shows an opened shaping tool according to FIG. **35** with a protective element;

[0247] FIG. **44** shows a schematic sectional representation of a protective element, arranged on a mass, with shielding deflection zone, concave zone and convex counterzone;

[0248] FIG. **45** shows a shielding layer in the assembly with a plastic layer;

[0249] FIG. **46** shows a schematic sectional representation of a detail of a protective element, arranged on a mass, with shielding deflection zone;

[0250] FIG. **47** shows a shielding layer in the layered assembly with a plastic layer;

[0251] FIG. **48** shows a possibility for incorporating a contacting element in a method for producing a protective element;

[0252] FIG. **49** shows a schematic representation of a detail of a protective element obtained according to the method of FIG. **48**;

[0253] FIG. **50** shows a possibility for incorporating a contacting element in a further method for producing a protective element;

[0254] FIG. **51** shows a schematic representation of a detail of a protective element obtained according to the method of FIG. **50**;

[0255] FIG. **52** shows a possibility for incorporating a contacting element in a further method for producing a protective element;

[0256] FIG. **53** shows a possibility for incorporating a contacting element in a further method for producing a protective element;

[0257] FIG. **54** shows a schematic representation of a detail of a protective element obtained according to the method of FIG. **53**;

[0258] FIG. **55** shows a possibility for incorporating a contacting element in a further method for producing a protective element;

[0259] FIG. **56** shows a schematic representation of a detail of a protective element obtained according to the method of FIG. **55**;

[0260] FIG. **57** shows a possibility for incorporating a contacting element in a further method for producing a protective element;

[0261] FIG. **58** shows a schematic representation of a detail of a protective element obtained according to the method of FIG. **57**;

[0262] FIGS. **59-60** show a possibility for incorporating two contacting elements in a further method for producing a protective element;

[0263] FIG. **61** shows a schematic representation of a detail of a protective element obtained according to the method of FIGS. **59-60**;

[0264] FIGS. **62-63** show a possibility for incorporating two contacting elements in a further method for producing a protective element;

[0265] FIG. **64** shows a schematic representation of a detail of a protective element obtained according to the method of FIGS. **62-63**;

[0266] FIGS. **65-66** show a possibility for incorporating a contacting element and a sleeve in a further method for producing a protective element;

[0267] FIG. **67** shows a schematic representation of a detail of a protective element obtained according to the method of FIGS. **65-66**;

[0268] FIGS. **68-69** show a possibility for incorporating two contacting elements in a further method for producing a protective element;

[0269] FIG. **70** shows a schematic representation of a detail of a protective element obtained according to the method of FIGS. **68-69**;

[0270] FIGS. **71-72** show a possibility for incorporating a contacting element in a further method for producing a protective element;

[0271] FIG. **73** shows a schematic representation of a detail of a protective element obtained according to the method of FIGS. **71-72**;

[0272] FIGS. **74-75** show a possibility for incorporating a contacting element in a further method for producing a protective element;

[0273] FIG. **76** shows a schematic representation of a detail of a protective element obtained according to the method of FIGS. **74-75**;

[0274] FIG. **77** shows a possibility for incorporating a contacting element in a further method for producing a protective element;

[0275] FIG. **78** shows a schematic representation of a detail of a protective element obtained according to the method of FIG. **77**;

[0276] FIGS. **79-80** show a possibility for incorporating a contacting element in a further method for producing a protective element;

[0277] FIG. **81** shows a schematic representation of a detail of a protective element obtained according to the method of FIGS. **79-80**;

[0278] FIG. **82** shows a schematic representation of a mold half with punch element;

[0279] FIG. **83** shows the schematic representation from FIG. **82** with contacting element;

[0280] FIG. **84** shows the schematic representation from FIG. **82** with contacting element;

[0281] FIG. **85** shows a schematic representation of a mold half with punch element and shielding material;

[0282] FIG. **86** shows a schematic representation of a mold half with punch element, shielding material and counterhold elements;

[0283] FIG. **87** shows the representation from FIG. **86** after production of a passage opening;

[0284] FIG. **88** shows the representation from FIG. **87** with finally reshaped shielding layer overhang;

[0285] FIGS. **89-91** show a possibility, building on the representations in FIGS. **82-88**, for integrating a contacting element in a further method for producing a protective element;

[0286] FIGS. **92-93** show a further possibility, building on the representations in FIGS. **82-88**, for integrating a contacting element in a further method for producing a protective element;

[0287] FIG. **94** shows a greatly simplified perspective representation of a protective element; and

[0288] FIG. **95** shows a section through the protective element from FIG. **94** with additional details.

DETAILED DESCRIPTION OF THE DRAWINGS

[0289] FIG. **1** shows a protective element **102** embodied as an aluminum component **100**. The protective element **102** comprises reinforcing ribs **104** which likewise consist of aluminum. The protective element **102** shown may be installed as a battery casing part **106**. In that case it may function in particular as an underbody protection element **108**.

[0290] The protective element **102** comprises an insulation layer **110**, which may consist, for example, of a foam material.

[0291] The protective element shown in FIG. **2** differs from the protective element shown in FIG. **1** in that it is not embodied as an aluminum component **100**, instead comprising a separate shielding component **109** and a further casing component **112**, with the shielding component **109** and the casing component **112** being separately fabricated and subsequently joined. The shielding component **109** may be fabricated, for example, from aluminum and may form a shielding layer **114**, which is able very largely to shield from electromagnetic radiation.

[0292] FIG. **3** shows a protective element **102**. The protective element **102** comprises a shielding layer **114**. The shielding layer **114** is an aluminum foil **116**. The protective element **102** comprises a plastic layer **118**. The plastic layer **118** completely covers the shielding layer **114**.

[0293] The protective element **102** shown in FIG. **3** comprises a contacting zone **120**. The contacting zone **120** is configured on a shielding layer end face. The shielding layer end face is an inner-surface contacting zone **122** encircling a passage opening. Electrical contact between the shielding layer **114** and a device on which the protective element may be arranged can be produced by means of the inner-surface contacting zone **122**.

[0294] The protective element **102** shown in FIG. **3** comprises reinforcing layers **124**. These are reinforcing layers **124** reinforced with fibers **126**. The reinforcing layers **124** reinforced with fibers

126 are organic sheets **128**.

[0295] The protective element shown in FIG. 3 comprises a protective layer **130**. The protective layer **130** is an insulation layer **110**. The insulation layer **110** may be electrically insulating and be an electrically insulating protective layer. Alternatively or additionally, the insulation layer may be a high-temperature thermal insulation layer.

[0296] Together with the contacting element **132**, the protective element **102** forms a component set **134**.

[0297] The contacting element **132** is a connecting element **133**. The contacting element **132** is in electrically conductive contact with the contacting zone **120**. The contacting element **132** is a screw **136**. The screw **136** comprises a thread **138**.

[0298] FIG. 4 shows a detail from FIG. 3 in enlarged representation. The contacting zone **120** is an inner-surface contacting zone **122** formed on the encircling shielding layer end face **140**.

[0299] The mass **142** shown in FIG. 4 forms a part of the device which emits electromagnetic radiation. In the case of a motor vehicle and also in the case of motorcycles and pedal cycles, the mass potential is the bodywork or the frame, respectively. The term “mass” refers to this mass potential. In FIG. 4, the mass **142** is a frame **146** which belongs to a carrier structure **144**. There is electrical contact between the shielding layer **114** and the frame **146** comprised by the device, by way of the contacting element **132**.

[0300] FIG. 5 shows section B-B from FIG. 3. The plastic layer **118** is a plastic core **148** which is arranged between reinforcing layers **124**. One of the reinforcing layers **124** forms a capping ply **150**. In the layered assembly, the capping ply **150** forms a capping layer **152**. The shielding layer **114** is arranged between the other reinforcing layer **124** and the insulation layer **110**, which forms a protective layer **130**. The insulation layer **110** represents a further capping ply **150**, which in the layered assembly forms a further capping layer **152**.

[0301] A capping layer herein refers to the layers which conclude in the layer structure of the layered assembly.

[0302] FIG. 6 shows a protective element **102**, comprising a contacting zone **120**, wherein the contacting zone **120** is an outer-surface contacting zone **154**.

[0303] FIG. 7 shows section A-A from FIG. 6. An outer surface **156** of the shielding layer **114** forms the outer-surface contacting zone **154**.

[0304] In the region of the protective element **102** shown by section A-A, the shielding layer **114** is wholly on the outside in the layer structure. It forms a capping layer **152**. The outer-surface contacting zone **154** is in direct electrical contact with the mass **142**. By means of the outer-surface contacting zone **154**, therefore, it is possible to produce electrical contact between the shielding layer **114** and the device which comprises the mass **142**.

[0305] FIG. 8 shows section B-B from FIG. 6. The layer structure shown in FIG. 8 corresponds to the layer structure shown in FIG. 5, as the protective elements from FIGS. 3 and 6 at the section point B-B have identical construction.

[0306] FIG. 9 shows a detail of a protective element **102** in the region of the contacting zone **120**. The protective element comprises an offset zone **158**. The shielding layer **114** extends in a different direction in the offset zone **158** than in the contacting zone **120**. In the contacting zone **120**, the shielding layer **114** forms a capping layer **152** of the protective element **102**. In an insulation zone **160**, which is separated from the contacting zone **120** by the offset zone **158**, an electrically insulating protective layer **130** forms the capping layer **152**.

[0307] The plastic layer **118** is thicker in a portion of the plastic layer **118** that overlaps with the contacting zone **120** than in a portion of the plastic layer **118** that overlaps with the insulation zone **160**. An adhesion promoter **164** is arranged on each of the surfaces of the shielding layer **114**. The contacting zone **120** is present at locations where the adhesion promoter **164** exhibits interruptions.

[0308] FIG. 9 as well shows a shielding layer end face **140**. There, however, the shielding layer end face **140** does not form an inner-surface contacting zone encircling a passage opening, as the thread

138 of the screw **136** is at a distance from the shielding layer end face **140**.

[0309] FIG. **10** shows a detail of a protective element **102** similar to the protective element **102** shown in FIG. **9**. The shielding layer end face **140** forms an inner-surface contacting zone **122**. Sleeve **166** forms a contacting element. There is electrical contact between the inner-surface contacting zone **122** and the contacting element **168**. There is also electrical contact by the sleeve **166** with the mass **142**. This electrical contact is promoted by the sleeve **166** being pressed by the screw **136** onto the mass **142**.

[0310] FIG. **11** shows a detail of a protective element. The protective element shown therein comprises an offset zone **158** and an insulation zone **160**. The protective element **102** additionally comprises contacting zones **120**. The contacting zones **120** comprise an outer-surface contacting zone **154** and an inner-surface contacting zone **122**. Electrical contact between the shielding layer **114** and the mass **142** comprised by the device is produced by means of these contacting zones. There is direct electrical contact of the outer-surface contacting zone **154** with the mass **142** and there is also contact mediated by the contacting element **132** between the inner-surface contacting zone **122** and the mass **142** comprised by the device. The contacting element is a connecting element **133** embodied as a screw **136**. The screw **136** is a flow-hole screw **170**.

[0311] FIG. **12** shows a detail of a protective element **102**. The protective element **102** comprises a protrusion zone **172**. The protrusion zone **172** protrudes from a surface of the protective element **102** and defines the contacting zone **120**. The contacting zone **120** is therefore an outer-surface contacting zone **154**.

[0312] In the protrusion zone **172**, the shielding layer **114** protrudes from the surface of the protective element **102**. In the protrusion zone **172**, the shielding layer **114** comprises a protrusion element **174**. The protrusion element **174** is a bead **176**, with the bead **176** protruding from the surface of the protective element. Thread **138** engages in a counterthread present in the mass **142**, and so the outer-surface contacting zone **154** defined by the protrusion zone **172** is pressed onto the surface of the mass **142**. In this process, the sealing element **180** arranged in the recess **178** is also compressed. The shielding layer end face **140** at a distance from the thread **138** does not form an inner-surface contacting zone.

[0313] FIG. **13** shows a portion of a protective element **102**. The contacting element **132** is a washer **182**. The contacting element **132** comprises protrusion elements **174** facing the shielding layer **114**. The contacting element **132** comprises protrusion elements **174** facing away from the shielding layer **114**.

[0314] The protrusion elements **174** mediate electrically conductive contacts of an outer surface of the shielding layer **114** with the contacting element **132** and of the contacting element **132** with the mass **142** when the protective element **102** is pressed onto the mass by way of the thread **138** of the screw **136** and the corresponding counterthread in the mass **142**. The protrusion elements **174** perforate the adhesion promoter **164**, thereby mediating the electrically conductive contact of the shielding layer via the contacting element with the mass. The contacting element **132** in the form of the washer **182** functions as a bridging element **184**.

[0315] FIG. **14** shows a detail of a protective element **102**. An adhesion promoter **164** is arranged on each of the surfaces of the shielding layer. The protective element **102** therefore does not comprise an outer-surface contacting zone **154**, but only an inner-surface contacting zone **122**. By means of the inner-surface contacting zone **122**, electrical contact is produced by the contacting element **132** between the shielding layer **114** and the mass **142** which is comprised by the device. The contacting element **132** is a connecting element **133**. The connecting element **133** is a screw **136** with a thread **138**.

[0316] FIG. **15** shows a detail of a protective element **102**. An inner-surface contacting zone **122** is configured on a shielding layer end face **140**. The protective element **102** does not comprise an outer-surface contacting zone **154**. An electrically insulating protective layer **130** completely covers the shielding layer **114**. The protective layer **130** forms the capping layer **152**, and so there is no

direct electrical contact between a surface of the shielding layer **114** and the mass **142** comprised by the device. The shielding layer **114** is therefore concealed on the contacting zone by an electrically insulating capping layer **152**.

[0317] The protective element **102** comprises an integrated electrically conductive discharge element **186** in the form of a sleeve **166**. The electrically conductive discharge element **186**, i.e. the sleeve **166**, is in electrically conductive contact with the inner-surface contacting zone **122**. The electrically conductive discharge element **186**, i.e. the sleeve **166**, extends through the electrically insulating capping layer **152**.

[0318] FIG. **16** shows an electrically conductive discharge element **186**. The discharge element **186** is a sleeve **166**. The discharge element **186** comprises an end-face surface **188**. The end-face surface comprises protrusion elements **174** which project from the end-face surface **188**.

[0319] Depending on the way in which the discharge element **186** is integrated into the protective element **102**, the discharge element **186** may serve as a contacting element **132**, as a connecting element **133** and/or as a bridging element **184**.

[0320] FIG. **17** shows a detail of a protective element **102**. The protective element **102** comprises an integrated electrically conductive discharge element **186** according to FIG. **16**. The electrically conductive discharge element **186** is in electrically conductive contact with the inner-surface contacting zone **122**. The electrically conductive discharge element **186** is also in electrically conductive contact with the mass **142** comprised by the device. The electrically conductive contact with the mass **142** is mediated by the protrusion elements **174**. Depending on the material of which the screw **136** is formed, there may be further electrically conductive contact by the inner-surface contacting zone **122** via the discharge element **186** and the screw **136** serving as connecting element **133**. The protective element **102** shown in FIG. **17** does not comprise an outer-surface contacting zone **154**, because an electrically insulating capping layer **152** completely covers the shielding layer.

[0321] FIG. **18** illustrates a shaping method **190** of the invention. The shaping method **190** is a compression molding method **192**.

[0322] FIG. **18** shows four different states which are traversed in the course of the compression molding method. The four states are each distinguished by a defined degree of reshaping and by a defined degree of mold filling. From left to right, the degree of reshaping, which increases in the direction of the arrow, is around 10%, around 20%, around 80%, and 100%. From left to right, the degree of mold filling, which increases in the direction of the arrow, is around 30%, around 30%, around 40%, and 100%.

[0323] In the state represented on the very left, the shaping tool **194** is open. The shaping tool **194** is a compression mold **196**. Provided in the open cavity **198** of the shaping tool **194** is an arrangement comprising the following: a ply of a shielding material **214**, a plurality of reinforcing plies **224**, and a plastic **218** arranged between at least two of the reinforcing plies. The shape of the protective element is defined at least partially by closing the shaping tool **194**, by building up an elevated pressure and/or by supplying heat. In this case, the shaping tool **194** passes through the four states represented in FIG. **18**.

[0324] In the state represented on the very right, the shape of the protective element **102** is defined nearly completely or completely and the cavity **198** is substantially completely closed.

[0325] Depending on the nature of the arrangement and particularly of the edge region of the arrangement, the protective element **102** may adopt very different structures on the contacting zone, these structures being described more closely in the subsequent FIGS. **19** to **26**.

[0326] FIG. **19** shows an edge portion of a resulting protective element **102**. FIG. **19** shows three respective reinforcing plies **224** on the two surfaces of the plastic melt **200** arising from the plastic **218**. The shielding material **214**, which is an aluminum foil **116**, is arranged between two of the reinforcing plies **224**. The shaping element **202** is a compression molding element **204** and forms a constituent of the shaping tool **194**, which is a compression mold **196**.

[0327] The arrow pointing vertically downward and drawn in on the shaping element **202** in FIG. **19** indicates a pressure which acts on the arrangement of plies represented. The effect of the pressure is to move the front **206** of the plastic melt **200** and to fill the space between the reinforcing plies.

[0328] FIG. **20** shows an edge portion of a further resultant protective element **102**. The resultant protective element **102** comprises an overlap portion **208** and an overhang portion **210**. In the overhang portion **210**, the shielding material **214**, i.e. the aluminum foil **116**, and a part of the reinforcing plies **224** extend beyond another part of the reinforcing plies **224**. The pressure built up by the compression molding element **204** produces an offsetting force **212**. The effect of the offsetting force **212** is to bend the shielding material **214** in the overhang portion. The bending is not represented in FIG. **20**. As a result, the development of an offset zone **158** may be produced in the compression molding step, such a zone being shown in FIG. **11**, for example. In the protective element **102** obtained on conclusion of the compression molding step, the shielding layer **114**, which originates from the shielding material **214**, has an outer-surface contacting zone **154** in the overhang portion **210**. This is because in the overhang portion **210**, the shielding layer **114** forms a capping layer **152**.

[0329] FIG. **21** shows an arrangement in which a reinforcing ply **224** arranged between shielding material **214** and the plastic melt **200** does not extend into the overhang portion **210**. The plastic melt **200** is therefore able to make direct contact with the shielding material **214** and produce the bending of the shielding material described in connection with FIG. **20**.

[0330] In the case of the arrangement shown in FIG. **22**, the shielding material **214** is arranged between the plastic melt **200** and a reinforcing ply **224**. In contrast to the arrangements shown in FIGS. **20** and **21**, there is no reinforcing ply between the plastic melt **200** and the shielding material **214**.

[0331] The arrangement shown in FIG. **23** corresponds to the arrangement shown in FIG. **19**. In contrast to FIG. **19**, the shielding material **214**, i.e. the aluminum foil **116**, is arranged on the very outside in the ply structure of the arrangement. The shielding material **214** forms a capping ply **150**. In the fully produced protective element **102**, therefore, there may be an outer-surface contacting zone **154**, without any need for an offsetting force described in connection with FIGS. **20** to **22** to bend the shielding material **214**.

[0332] The arrangement shown in FIG. **24** corresponds to the arrangement from FIG. **23**. In contrast to the arrangement in FIG. **23**, the shielding material **214**, i.e. the aluminum foil **116**, is arranged on the other side of the arrangement in the ply structure.

[0333] FIGS. **25** and **26** illustrate the state respectively before and after the bending of the shielding material **214** as a result of the offsetting force **212**. Clearly apparent in FIG. **26** is an offset zone **158** which has come about, as a consequence of the bending, at the transition between overhang portion **210** and overlap portion **208**.

[0334] FIG. **26** as well shows the contacting zone **120** which is formed in the completed protective element **102** and is an outer-surface contacting zone **154**.

[0335] On conclusion of the compression molding illustrated in FIGS. **25** and **26**, a protective element **102** may be produced, the edge of which is shown in FIG. **27**. In the example shown therein, the shielding layer **114** has two approximately right-angled bends in the region of the offset zone **158**. On one side of the offset zone, the reinforcing layer **124** forms the capping layer **152**. On the other side of the offset zone, the shielding layer **114** forms the capping layer **152**.

[0336] FIG. **28** shows section A-A through the protective element **102** shown in FIG. **27**.

Additionally represented is the mass **142** comprised by the device. In this region, the reinforcing layer **124** forms the capping layer **152**. The latter is arranged between the mass **142** and the shielding layer **114**. There is consequently no contacting zone **120** there.

[0337] FIG. **29** shows section B-B through the protective element **102** shown in FIG. **27**. The shielding layer **114** forms the capping layer **152**. There is electrical contact between the shielding

layer **114** and the mass **142** comprised by the device. The mass may be formed by a frame **146** of a carrier structure **144**. In the region of the section B-B, therefore, the protective element comprises a contacting zone **120**. The contacting zone is an outer-surface contacting zone **154**.

[0338] The arrangement shown in FIG. **30** corresponds to the arrangement shown in FIG. **25**. In the case of the arrangement shown in FIG. **30**, however, only the shielding material **214** extends into the overhang portion **210**. In the example shown here, the offset zone **158** is a preformed offset zone **158**, which is prelocated in the shielding material **214** which is introduced into the shaping tool **194**.

[0339] Especially if only the shielding material **214** extends into the overhang portion **210**, it may be advantageous to hold overhanging shielding material **214** in position with the aid of a securement arrangement **220** and to form bends characteristic of the offset zone in the shielding material **214**. This is indicated in FIG. **30** in the section represented on the left.

[0340] FIGS. **31** and **32** show that during compression molding, the plastic melt **200** enters the overhang portion **210**.

[0341] FIG. **33** shows section A-A through the protective element **102** shown in FIG. **32**, with the mass **142** being represented additionally. There, the reinforcing layer **124** forms the capping layer **152**. There is no contacting zone **120** there, as the reinforcing layer is arranged between the shielding layer and the mass comprised by the device.

[0342] FIG. **34** shows section C-C through the protective element **102** represented in FIG. **32**, with the mass **142** being represented additionally. There, there is electrical contact between the shielding layer **114** and the mass **142** comprised by the device. The protective element therefore comprises a contacting zone **120**. The contacting zone **120** is an outer-surface contacting zone **154**. In the region of the contacting zone **120**, the protective element **102** does not comprise a reinforcing layer **124**; other than the shielding layer **114**, the layer structure contains only the plastic layer **118**.

[0343] FIGS. **35** to **43** illustrate a shaping method for producing a protective element **102**. The shaping method is an injection molding method.

[0344] The injection molding method takes place in an injection mold **230**. The injection mold **230** comprises a first mold half **232** and a second mold half **234**.

[0345] The first mold half **232** comprises a first feed zone **236** for the supply of a plastic melt **200**. The first mold half **232** comprises guiding zones **238**. In the guiding zones **238**, retaining elements **240** are guided. The retaining elements **240** can be used for punching. These are punching elements **242**.

[0346] The second mold half comprises a second feed zone **244**. The second mold half comprises retaining elements **240** guided in guiding zones **238**. The retaining elements **240** can be used for punching. These are punching elements **242**.

[0347] FIG. **35** shows the injection mold in an opened state with open cavity **198**.

[0348] FIG. **36** shows the positioning of a ply of a shielding material **214** between the mold halves **232** and **234**. For positioning the shielding material **214** between the mold halves **232** and **234**, the injection mold **230** can be partially closed and/or retaining elements **240** guided in the guiding zones **238** can be moved toward one another. In particular, a respective retaining element **240** of a mold half **232**, **234** may lie against edges **215** of the shielding material such that the shielding material **214** is held in a desired position by the retaining elements **240**. This is shown in FIG. **37**.

[0349] By further closing of the injection mold **230**, the shielding material can be reshaped and hence a shape of the shielding material **214** that is desired for the protective element **102** can be defined in the injection mold **230**. FIG. **38** shows the injection mold **230** and the shielding material **214** which is located therein and has been brought to the desired shape.

[0350] A plastic layer **118** may be applied to one of the two surfaces of the shielding material **214**. For this purpose, the plastic melt **200** may be conveyed from the first feed zone **236** into a gap which exists between one of the surfaces of the shielding material **214** and a surface of the first mold half **232**. In the largely closed injection mold **230**, the retaining elements **240** and the

reshaped shielding material held by the retaining elements may be moved toward the second mold half **234** for this purpose. Accordingly, the width of the gap and hence ultimately the thickness of the plastic layer **118** to be applied can be controlled. The processing state attained after the application of the plastic layer **118** is represented in FIG. **39**.

[0351] The injection mold may be easily opened by modification to the position of the second mold half **234**, producing a gap between the surface of the second mold half **234** and the shielding layer **114**. Plastic melt **200** may be passed through the second feed zone **244** into the gap, so that a plastic layer **118** is also applied to the other of the two surfaces of the shielding material **214**. In this case, the fabrication state represented in FIG. **40** is attained.

[0352] Through both feed zones **236** and **244**, additional plastic melt **200** may be pressed into the two gaps and/or the injection mold may be further closed. In this way it can be ensured that edge zones of the two gaps as well are filled up completely with plastic melt **200**, as shown by FIG. **41**.

[0353] The position of the retaining elements in the guiding zones may be modified. Regions of the shielding material **214** that are arranged between the retaining elements and are not covered with a plastic layer **118** can be separated by punching. This is illustrated by FIG. **42**.

[0354] The injection mold **230** can be opened and the resulting protective element **102** withdrawn (FIG. **43**).

[0355] FIGS. **44** to **47** illustrate protective elements **102**, wherein the shielding layer **114** comprises a shielding deflection zone **246**. In the shielding deflection zone **246**, two portions **248** of the shielding layer **114** overlap. One of the two portions **248** forms the contacting zone **120**. The contacting zone **120** is an outer-surface contacting zone **154**. This is illustrated in particular by FIG. **44** and FIG. **46**.

[0356] The protective element **102** shown in FIG. **44** differs from the protective element shown in FIG. **46** in particular in that one of the two overlapping portions **248** of the shielding layer **114** extends along a concave zone **250** of the protective element **102**, and the other of the two overlapping portions **248** of the shielding layer extends along a convex counterzone **252** of the protective element **102**. The convex counterzone **252** is arranged in the concave zone **250**.

[0357] The production of the protective elements **102** with shielding deflection zone **246** which are shown in FIGS. **44** and **46** is illustrated in FIGS. **45** and **47**. A layered assembly can be provided, comprising a shielding layer and a plastic layer. The layered assembly may have a cutout zone **254**, in which the plastic layer may for example be embodied with interruption or as substantially thinner. The cutout zone **254** may enable the shielding layer **114** to be bent round, that is may enable the production of a shielding deflection zone. The shielding layer is preferably bent round with an approximate U shape in the cutout zone. If the concave zone **250** and the convex counterzone **252** are configured at a suitable distance from the cutout zone **254**, the convex counterzone **252** can come to lie in the concave zone **250** when the shielding layer **114** is bent round. This becomes clear from FIG. **45**.

[0358] FIGS. **48-93** illustrate diverse possibilities for forming contacting zones **120** through the additional use of contacting elements **132** in a shaping method which may be carried out, for example, according to the principle shown in FIG. **18** or else according to the principle shown in FIGS. **35-43**. In this case, modified mold halves **232** and **234** may be used. These mold halves may have die elements **241** and/or punch elements **262** guided in guiding zones **238**. Alternatively or additionally, the modified mold halves **232** or **234** may have recesses for accommodating contacting elements **132**.

[0359] FIGS. **48, 52, 53, 55, 57, 59, 60, 62, 63, 65, 66, 68, 69, 71, 72, 74, 75, 77, 79** and **80** in particular show how, in order to incorporate a contacting element **132** in methods for producing a protective element **102**, a die element **241** can be guided in a guiding zone **238** or a contacting element **132** can be accommodated in a recess in a mold half **232** or **234**.

[0360] FIGS. **82-93** show how a punching element **262** guided in a guiding zone **238** may be used for introducing a passage opening **121** into a shielding material **214**.

[0361] With all of the possibilities that are illustrated in FIGS. **48-93**, it is possible, during the formation of a plastic layer **118**, for the annular contacting element **132**, which may be a sleeve **166**, for example, to be embedded at least partially into the resultant plastic layer **118**. In that case, the contacting element **132** is arranged, for example, on a surface of the shielding material **214**. The contacting element **132** here may be positioned at a desired point using the die element **241** or the punch element **262** (see, in particular, FIGS. **48, 53, 55, 57, 59, 60, 62, 63, 65, 66, 68, 69, 88-93**) or accommodated partially in a recess in a mold half **232** or **234** (see, in particular, FIGS. **71, 74, 79**) or anchored in a layer, e.g. plastic layer **118**, formed in a preceding method step (see, in particular, FIGS. **72, 75, 80**). The plastic melt **200** is then conveyed in that direction such that the front **206** of the spreading plastic melt **200** is able to spread over the contacting element **132**. This allows the contacting element **132** to be embedded into the resultant plastic layer **118**.

[0362] In the case of the method illustrated in FIG. **48**, the die element **241** comprises a shoulder **245** and an extent element **243** which extends beyond the shoulder **245**. The contacting element **132** is taken up onto the extent element **243**. A diameter of the extent element **243** is adapted to the internal diameter of the contacting element **132**. At an end-face surface **188**, the contacting element **132** comprises a plurality of protrusion elements **174**. The contacting element **132** is held with the die element **241** on the surface of the shielding material **214**. The contacting element **132** is pressed with the die element **241** against the surface of the shielding material **214**. The protrusion elements **174** are impressed into the surface of the shielding material **214**. In the resulting protective element, of which a detail is shown in FIG. **49**, the tips of the protrusion elements **174** respectively form a protrusion element overhang **175**, which penetrates the shielding layer **114** and is able to reinforce or stabilize contact with a device, as a contact-reinforcing element **177**.

[0363] FIG. **50** illustrates a possibility for producing the protective element **102** shown in FIG. **51**. A die element **241** guided in a guiding zone **238** occupies a space here, around which the plastic melt **200** spreads. In a downstream step, the contacting element **132** can be introduced into the space and the shielding material **214** can be pierced with the protrusion elements **174** of said contacting element. In contrast to the protective element shown in FIG. **49**, the contacting element **132** in the case of the protective element **102** shown in FIG. **51** is not materially bonded with the plastic layer **118**. It is alternatively possible, indicated in FIG. **52**, to bend an initially formed shielding layer overhang **179** into the space made with the die element **241**, allowing an outer-surface contacting zone **154** to come into contact with an outer lateral surface of the contacting element **132** when the contacting element **132** is introduced into the space.

[0364] If the contacting element **132** and the die element **241** have shoulders **245** matched with an accurate fit to one another, a shielding material **214** can be introduced between the die element **241** and the contacting element **132** and reshaped between the shoulders **245**. Additionally, the die element **245** can be pressed on so firmly that the protrusion elements **174** pierce the shielding material **214** (FIGS. **53** and **54**). Arranged on a shoulder **245** of the die element **241**, there may be an auxiliary ring element **260** which, after the die element **241** has been pressed onto the shielding material **214**, can come to lie on a shoulder **245** of the contacting element **132** that is lined with the shielding material (FIGS. **55** and **56**).

[0365] FIG. **57** shows a possibility wherein a die element **241** can have a die recess **249** which may be radially outwardly limited by an annular projection **251**. A shielding material **214** may extend through the die recess **249**. This material can be clamped between a contacting element **132** and the die element **241** in the die recess **249** and reshaped by a movement of the die element **241**. After the plastic layer **118** has been attached, a contacting zone **120** of the shielding layer **114** is in electrically conductive contact with an outer lateral surface of the contacting element **132**. The contacting zone **120** is an outer-surface contacting zone **154**. In the region of the contacting zone, the shielding layer **114** is pressed through the plastic layer **118** onto the surface of the contacting element **154** (FIG. **58**).

[0366] FIGS. **59-70** show possibilities for producing protective elements **102** which comprise at

least one contacting element **132** respectively on both sides of the shielding layer **114**.

[0367] In this case, on one side of a shielding material **214** in each case, a first contacting element **132** is held by a die element **241**, which engages into the contacting element **132**, and then a plastic melt **200** is spread flatly around the contacting element **132** such that a material bond can be produced between the resultant first plastic layer **118** and the contacting element **132**.

[0368] Moreover, on the other side of the shielding material **214**, a second contacting element **132** is held by a die element **241**, which engages into the contacting element **132**, and then a further plastic melt **200** is spread flatly around the second contacting element **132** such that a material bond can be produced between the resultant second plastic layer **118** and the contacting element **132**.

[0369] Here, on an end-face surface **188** of a contacting element **132**, a shoulder and an annular projection **251** may be configured such that the shielding material **214** can be clamped between the annular projection **251** and the other contacting element **132** (see FIGS. **62-64**).

[0370] One of the contacting elements **132** may be radially external. The other contacting element **132**, which may also be a sleeve **166** which need not be electrically conductive, may be radially internal. A shielding layer overhang may extend into an intervening gap (FIGS. **65-67**).

[0371] The shielding layer **114** may lie between the mutually facing end-face surfaces of the two contacting elements **132**. Protrusion elements **174** which may be arranged on the mutually facing end-face surfaces of both contacting elements **132** may preferably be pressed onto the shielding layer **114** or penetrate the shielding layer **114**. The contacting elements **132** may comprise flange zones **256**. The protrusion elements **174** may be configured on the flange zones **256** (FIGS. **68-70** and FIG. **16**).

[0372] FIGS. **71** and **72** illustrate a possibility for producing the protective element **102** shown in FIG. **73**. The mold half **234** comprises a recess in which a part of the contacting element **132** is accommodated (FIG. **71**). The recess is matched to the shape of the contacting element **132**, which on its outer lateral surface comprises a radially outwardly extending crown ring zone **258**. It is evident from FIG. **71** that the crown ring zone **258** is likewise accommodated in the recess.

[0373] A first plastic layer is produced around the contacting element **166** between the shielding material **214** and the mold half **232** shown at the bottom in FIG. **71**. A second plastic layer is subsequently produced around the contacting element **166** between the shielding material **214** and the mold half **24** shown at the top in FIG. **72**.

[0374] FIGS. **74** and **75** illustrate a possibility for producing the protective element **102** shown in FIG. **76**. The procedure is as shown in FIGS. **71** and **72** and is described in association with these two figures; however, the recess is not configured for accommodating the crown ring zone **258**, and the shielding material **214** comes to lie on the other side of the crown ring zone **258**.

[0375] FIG. **77** illustrates a possibility for producing the protective element **102** shown in FIG. **78**. A shielding layer overhang **179** is arranged on an outer lateral surface of the contacting element **132**. The plastic melt **200** is then spread around the contacting element **132**, so that the resulting plastic layer **118** presses a contacting zone **120**, which is configured on the shielding layer overhang **179** and is an outer-surface contacting zone **154**, onto the outer lateral surface and is consequently able to promote or stabilize electrical contact.

[0376] FIGS. **79** and **80** illustrate a possibility for producing the protective element **102** shown in FIG. **81**. The procedure in this case is as described in connection with FIGS. **77** and **78**. However, a longer contacting element **132** is used, which is first accommodated in a recess in a mold half **234** and after the production of the first plastic layer **118** is accommodated by a second plastic melt **200**.

[0377] FIG. **82** shows a punch element **262** guided in a guiding zone **238** of a mold half **234**. The punch element **262** comprises an extent element **243**.

[0378] FIG. **83** shows the punch element **262** guided in the guiding zone **238** of the mold half **234**, with the extent element **243** extending through a contacting element **132** which is a sleeve **166**.

[0379] FIG. **84** corresponds to the representation from FIG. **83**, with the punch element **262** being represented in a different position in the guiding zone. The tip of the punch element does not

protrude from the guiding zone **238**.

[0380] FIG. **85** corresponds to the representation from FIG. **84**. Additionally shown is a shielding material **214** lying against the mold half **234**.

[0381] Additionally to FIG. **85**, FIG. **86** also shows a further mold half **232**. The latter comprises guiding zones **238** for counterhold elements **264**. The counterhold elements **264** are able to press the shielding material **214** against the mold half **234** against which the shielding material **214** is lying.

[0382] FIG. **87** shows the formation of a passage opening **121** in the shielding material **214**. The passage opening **121** comes about by the punch element **262** in the guiding zone **238** being moved up to the shielding material **214** and, with its tip, being moved through the shielding material. This results in a shielding layer overhang **179** on one edge of the passage opening **121**.

[0383] FIG. **88** shows that on continuation of the movement of the punch element **262**, the contacting element **132** is guided partially through the passage opening **121**. The contacting element sits on a surface of the mold half **232**. A shielding layer overhang **179** is arranged here on an outer lateral surface of the contacting element **132**.

[0384] FIG. **89** shows how subsequently the plastic melt **200** is spread around the contacting element **132**.

[0385] FIG. **90** shows the spreading of a second plastic melt **200** on the other side of the shielding material **114**, with the state shown in FIG. **91** being attained.

[0386] FIGS. **92** and **93** show a possibility for producing the second plastic layer **118** with a different mold half **234**.

[0387] FIG. **94** shows a protective element **102**, in a greatly simplified representation; said protective element may be, for example, a casing tray **284** for an electrochemical energy storage device of a motor vehicle, for example for a high-voltage battery device.

[0388] In FIG. **94**, a contacting zone **120** and a reinforcing unit **266** of the protective element **102** are also shown.

[0389] FIG. **95** shows a section of the protective element **102** from FIG. **94** along the line XCV-XCV, with further details being shown in the section.

[0390] The contacting element **132** represented additionally in FIG. **95** is deformable and elastic. The contacting element **132** comprises fibers **268** which are contained in a fibrous material **270**.

[0391] The fibrous material **270** of the contacting element **132** extends around a compression zone **272**. During assembly of the protective element **102** on a motor vehicle, the contacting element **132** may be deformed, in which case the contacting element **132** and its compression zone **272** are compressed, for example squashed, in particular in the assembly direction.

[0392] The contacting element **132** shown is a hollow contacting element **274** through which the compression zone **272** extends. The contacting element **132** may be, for example, a fiber material hose **278**, e.g. a woven fabric hose **276**.

[0393] The protective element **102** comprises a positioning and/or aligning element **280**, which in the example shown here is a positioning and/or aligning projection **282**. The positioning and/or aligning element **280** is arranged on the contacting zone **120**.

[0394] The fiber material hose **278**, e.g. the woven fabric hose **276**, is arranged on the positioning and/or aligning element **280**, e.g. on the positioning and/or aligning projection **282**, and on the contacting zone **120**.

LIST OF REFERENCE SIGNS

[0395] **100** aluminum component [0396] **102** protective element [0397] **104** reinforcing rib [0398]

106 battery casing part [0399] **108** underbody protection element [0400] **109** shielding component

[0401] **110** insulation layer [0402] **112** casing component [0403] **114** shielding layer [0404] **116**

aluminum foil [0405] **118** plastic layer [0406] **120** contacting zone [0407] **121** passage opening

[0408] **122** inner-surface contacting zone [0409] **124** reinforcing layer [0410] **126** fibers [0411] **128**

organic sheet [0412] **130** protective layer [0413] **132** contacting element [0414] **133** connecting

element [0415] **134** component set [0416] **136** screw [0417] **138** thread [0418] **140** shielding layer end face [0419] **142** mass [0420] **144** carrier structure [0421] **146** frame [0422] **148** plastic core [0423] **150** capping ply [0424] **152** capping layer [0425] **154** outer-surface contacting zone [0426] **156** outer surface [0427] **158** offset zone [0428] **160** insulation zone [0429] **164** adhesion promoter [0430] **166** sleeve [0431] **168** contacting element [0432] **170** flow-hole screw [0433] **172** protrusion zone [0434] **174** protrusion element [0435] **175** protrusion element overhang [0436] **176** bead [0437] **177** contact-reinforcing element [0438] **178** recess [0439] **179** shielding layer overhang [0440] **180** sealing element [0441] **182** washer [0442] **184** bridging element [0443] **186** discharge element [0444] **188** end-face surface [0445] **190** shaping method [0446] **192** compression molding method [0447] **194** shaping tool [0448] **196** compression mold [0449] **198** cavity [0450] **200** plastic melt [0451] **202** shaping element [0452] **204** compression molding element [0453] **206** front [0454] **208** overlap portion [0455] **210** overhang portion [0456] **212** offsetting force [0457] **214** shielding material [0458] **215** edge [0459] **218** plastic [0460] **220** securement arrangement [0461] **224** reinforcing ply [0462] **230** injection mold [0463] **232**, **234** mold half [0464] **236**, **244** feed zone [0465] **238** guiding zone [0466] **240** retaining element [0467] **241** die element [0468] **242** punching element [0469] **243** extent element [0470] **245** shoulder [0471] **246** shielding deflection zone [0472] **247** transition zone [0473] **248** portion [0474] **249** die recess [0475] **250** concave zone [0476] **251** annular projection [0477] **252** convex counterzone [0478] **254** cutout zone [0479] **256** flange zone [0480] **258** crown ring zone [0481] **260** auxiliary ring element [0482] **262** punch element [0483] **264** counterhold element [0484] **266** reinforcing unit [0485] **268** fibers of the contacting element [0486] **270** fibrous material [0487] **272** compression zone [0488] **274** hollow contacting element [0489] **276** woven fabric hose [0490] **278** fiber material hose [0491] **280** positioning and/or aligning element [0492] **282** positioning and/or aligning projection [0493] **284** casing tray

Claims

1. A protective element for arrangement on a device which emits electromagnetic radiation or an at least partially electrically powered motor vehicle, the protective element comprising: a shielding layer for shielding from electromagnetic radiation, a plastic layer; and a contacting zone by means of which electrical contact can be produced between the shielding layer and the device or said at least partially electrically powered motor vehicle.
2. The protective element as claimed in claim 1, wherein the shielding layer is arranged between the plastic layer and an electrically insulating protective layer.
3. The protective element as claimed in claim 1, wherein the protective element comprises a contacting element, wherein by means of the contacting element electrical contact can be producible or stabilizable between the contacting zone and the device and/or wherein electrical contact bridged by the contacting element can be producible between the contacting zone and the device.
4. The protective element as claimed in claim 3, wherein the contacting element is deformable and/or elastic.
5. The protective element as claimed in claim 3, wherein the shielding layer comprises a shielding layer overhang, wherein the shielding layer overhang comprises the contacting zone, and wherein the shielding layer on the shielding layer overhang is not covered by the plastic layer and/or, if the protective element is embedded between the plastic layer and a further layer or a second plastic layer is not covered by at least one of these two layers.
6. The protective element as claimed in claim 3, wherein at least part of the shielding layer overhang extends in a different direction than a part of the shielding layer that is covered by the plastic layer, and wherein at least part of the shielding layer overhang is in contact with an outer lateral surface of the annular contacting element.

7. The protective element as claimed in claim 1, wherein the contacting zone is configured on a shielding layer end face.
8. The protective element as claimed in claim 3, wherein the shielding layer comprises a passage opening, and/or wherein the passage opening is able to align with an opening in the annular contact element.
9. The protective element as claimed in claim 1, wherein the contacting zone is an outer-surface contacting zone, and wherein the outer-surface contacting zone is configured on an outer surface of the shielding layer.
10. The protective element as claimed in claim 9, wherein the protective element comprises: an offset zone, wherein the shielding layer extends in a different direction in the offset zone than in the contacting zone.
11. The protective element as claimed in claim 9, wherein, in the contacting zone, the shielding layer forms a capping layer of the protective element.
12. The protective element as claimed in claim 10, wherein, in a further zone, separated from the contacting zone by the offset zone or in an insulation zone a layer of the protective element different from the shielding layer or the electrically insulating protective layer, forms a capping layer of the protective element.
13. The protective element as claimed in claim 1, wherein the protective element is fiber-reinforced.
14. The protective element as claimed in claim 13, wherein the plastic layer comprises fibers.
15. The protective element as claimed in claim 14, wherein the fibers are dispersed in a plastic material of the plastic layer, and wherein a mean length of the fibers is 1 mm to 40 mm or 2 mm to 20 mm.
16. The protective element as claimed in claim 1, wherein the protective element comprises a reinforcing layer, and wherein the reinforcing layer comprises a laid fiber scrim or a uniaxial or a multiaxial scrim or a woven fiber fabric.
17. The protective element as claimed in claim 1, wherein the protective element is a protective sandwich element, wherein the protective sandwich element comprises a reinforcing layer respectively on both surfaces, and/or wherein the two reinforcing layers independently of one another comprise a laid fiber scrim or a uniaxial or a multiaxial scrim or a woven fiber fabric.
18. A component set for arrangement on the device which emits electromagnetic radiation or the at least partially electrically powered motor vehicle, the component set comprising: the protective element as claimed in claim 1; and the contacting element which is in electrically conductive contact with the contacting zone.
19. A method for producing the protective element as claimed in claim 1, wherein a ply of a shielding material is provided and a plastic layer is formed in the layered assembly with the shielding material, wherein the shielding material is at least partially reshaped prior to the formation of the plastic layer; and/or is at least partially reshaped during the formation of the plastic layer; and/or is at least partially reshaped after the formation of the plastic layer, in the layered assembly, wherein a second layer is formed in the layered assembly with the shielding material concurrently with the formation of the plastic layer, temporally overlapping the formation of the plastic layer or temporally offset from the formation of the plastic layer, and/or wherein a contacting zone is formed by means of which electrical contact is produced between the shielding layer formed from the shielding material and the device which emits electromagnetic radiation.
20. The method as claimed in claim 19, wherein the method comprises a compression molding method, wherein a) an arrangement comprising: the ply of the shielding material, a second ply serving to form the second layer or a reinforcing ply, and a plastic arranged between these plies and serving to form the plastic layer or a plastic which is moldable by compression molding, is provided in an open cavity of a shaping tool, and b) a shape of the protective element is at least partially defined by closing the shaping tool, by building up an elevated pressure and/or by

supplying heat, wherein the shielding material is at least partially reshaped, and c) the shaping tool is opened and the protective element or a protective element precursor is removed from the open cavity of the shaping tool.

21. The method as claimed in claim 19, wherein the method comprises an injection molding method, wherein a) the ply of the shielding material is positioned between two mold halves of an injection mold, b) a shape of the shielding material, desired for the protective element, is defined in the injection mold by reshaping, c) the plastic layer or a layer of melted plastic, is applied on at least one of the two surfaces of the shielding material, and d) a plastic layer or the layer of melted plastic, is applied on the other of the two surfaces of the shielding material.
